Response to Reviewer Y2rf

Structure-Aware Prediction of PROTAC-Mediated Protein Degradability via Graph Neural Networks

ACM-BCB 2026 Submission

Rating: 2 — Reject

We appreciate the reviewer's detailed and rigorous critique. We have addressed every major and minor concern and believe the revised manuscript warrants reconsideration. All changes are highlighted in blue in the revised paper. Line numbers below refer to the revised `paper.tex`.

M1: The main result should be multi-seed average, not best seed. Rewrite abstract, results, and conclusion accordingly.

Response: Done. Throughout the revised manuscript, the multi-seed average is the primary reported result and the best seed is secondary:

- **Abstract** (line 65): Leads with "0.646±0.124 AUROC on target-unseen evaluation (best seed: 0.7449)"—average first, best seed parenthetical. [Paper reference: Line 65.]
- **Abstract closing** (line 68): "However, the high seed variance (std = 0.124) and limited E3 coverage require ensembling for reliable deployment." [Paper reference: Line 68.]
- **Introduction** (lines 132–133): Reports both 3-seed (0.646±0.124) and 6-seed (0.603±0.097) means as primary numbers; notes statistical non-significance ($p = 0.556$) and VHL failure (0.396) immediately. [Paper reference: Lines 132–133.]
- **Table 1** (lines 421–438): Now contains both 3-seed and 6-seed rows, with "Best Seed" as a separate column rather than the headline number. [Paper reference: Lines 421–438 (Table 1, `tab:main`).]
- **Results text** (lines 440–442): "The improved model achieves a 3-seed mean of 0.646... and a more conservative 6-seed mean of 0.603... The gradient boosting baseline achieves 0.607; the 3-seed advantage (+6.4%) is not statistically significant (p = 0.259)... The best seed (0.7449, +23% over baseline) reflects favorable initialization and should not be expected reliably." [Paper reference: Lines 440–442.]
- **Conclusion** (line 749): "On target-unseen evaluation, the 3-seed mean is 0.646±0.124 (best: 0.7449) and the 6-seed mean is 0.603±0.097; neither is statistically superior to gradient boosting (0.607)..." [Paper reference: Lines 749–750.]

M2: Target-unseen performance is unstable. Report more seeds and provide confidence intervals. Discuss average performance rather than peak.

Response: We trained 3 additional seeds (789, 1011, 1213), yielding AUROCs of 0.657, 0.520, and 0.502 respectively. The 6-seed results are:

Estimate	AUROC	95% CI
3-seed mean	0.646 ± 0.124	[0.506, 0.745]
6-seed mean	0.603 ± 0.097	[0.532, 0.682]
Gradient Boosting	0.607	—

The 6-seed mean (0.603) is marginally *below* the gradient boosting baseline (0.607, bootstrap $p = 0.556$). We report this honestly and prominently:

> *"This result is important to report honestly: the original 3-seed mean (0.646) was moderately favorable; the 6-seed analysis demonstrates that DegradoMap's advantage over hand-crafted feature baselines is not robust across all initializations."*

We explicitly recommend ensembling across $\geq$6 seeds for deployment (line 598) and note that individual seeds span a 0.25-AUROC range (0.502–0.745).

[Paper reference: Lines 593–599 ("Six-seed extended validation" paragraph in §4.3); Table 1 lines 421–438 (6-seed row added); Figure 4 at line 605 (all 6 seeds visualized); Seed sensitivity paragraph lines 590–592.]

M3: The lysine-related biological claim is not supported by ablation—removing the lysine indicator *improves* performance. Explain or retract.

Response: We now discuss this "lysine indicator paradox" directly in the Discussion (§5) with concrete overfitting evidence:

- With lysine indicator: val 0.584, test 0.477 (val-test gap: $-0.106 \rightarrow$ **overfitting**).
- Without lysine indicator: val 0.541, test 0.578 (val-test gap: $+0.038 \rightarrow$ generalizes).

This is a classic overfitting pattern: the lysine indicator improves validation but hurts test performance. We propose three explanations:

1. The lysine-weighted *pooling mechanism* (Eq. 3, line 247) already encodes lysine information architecturally, making the binary node feature redundant and causing overfitting to training-set lysine distributions.
2. The model learns that not all lysines are equal—the binary indicator oversimplifies, while the pooling weights discriminate based on structural context (SASA, disorder, local geometry).
3. The indicator may introduce a confounding bias: training targets have specific lysine distributions that don't generalize to unseen proteins.

We conclude: "architectural inductive biases (lysine pooling) are more effective than explicit feature annotation for this task," and note that current feature design is suboptimal.

[Paper reference: Lines 719–726 ("The lysine indicator paradox" paragraph in §5); Figure 7 at line 729 (lysine paradox visualization); Limitation item 7 at line 738 ("Lysine indicator counterproductive").]

M4: The E3 generalization claim should be narrower. Evidence supports mainly CRBN↔VHL transfer.

Response: Agreed and revised throughout:

- **Abstract** (line 65): Changed to "CRBN→VHL E3-unseen transfer." [Paper reference: Line 65.]
- **Introduction** (line 133): "VHL-targeted proteins fail below random (0.396 AUROC)—both limitations prominently acknowledged." [Paper reference: Line 133.]
- **Section heading** (line 608): "E3 generalization is limited to major ligases." [Paper reference: Line 608.]
- **E3-unseen vs target-unseen paradox explained** (lines 608–611): CRBN→VHL E3-unseen (0.811) succeeds because it tests E3-level transfer; VHL target-unseen (0.396) fails because it tests protein-level generalization—different axes. [Paper reference: Lines 608–611.]
- **VHL root cause analysis** (lines 612–620): Ruled out class imbalance (36.9% vs 38.4%) and dataset size (91 vs 125 proteins); identified structural heterogeneity as the likely cause. Explicit warning: "practitioners should expect substantially lower performance than the overall 0.646 mean when deploying on VHL-targeted proteins, and near-random performance for rare E3s." [Paper reference: Lines 612–620.]
- **Per-E3 Table 5** (lines 622–637): Quantitative breakdown showing CRBN 0.758, VHL 0.396, cIAP1 0.417. [Paper reference: Table 5 at line 625.]
- **Per-E3 Figure 5** (lines 639–645): Visual comparison of per-E3 performance. [Paper reference: Figure 5 at line 644.]
- **Conclusion** (line 749): "CRBN→VHL transfer (0.811 AUROC)." [Paper reference: Line 749.]

M5: Validation setup should better align with the target-unseen setting, or explain why the current setup is sufficient.

Response: We now explicitly acknowledge this as a methodological limitation in the Training Procedure section (§3.5):

> *"Validation-split AUROC does not reliably predict target-unseen test AUROC... Ideally, we would use a target-unseen validation split for hyperparameter selection; however, this would further reduce the already-limited training data (155 proteins → ∼120 training, ∼15 validation, ∼20 test)."*

We state that the chosen hyperparameters (LR=5×10^{-4}, dropout=0.05) may not be optimal for target-unseen generalization, and that this likely contributes to the observed variance.

[Paper reference: Lines 314–318 ("Validation-test mismatch" paragraph in §3.5 Training); Line 310 (hyperparameter values noted); Limitation item 4 at line 735 ("Validation-test mismatch").]

Minor 1: Dataset limitations should be stated more directly in main results.

Response: A "Dataset characteristics and limitations" paragraph now appears at the very start of the Results section (§4), *before* any performance numbers are presented. It states:

- Only 155 unique protein targets (effective diversity for target-unseen).
- E3 distribution: CRBN 62%, VHL 35%, remaining 8 E3s combined 3%.
- These constraints fundamentally bound generalization performance.

[Paper reference: Lines 412–417 ("Dataset characteristics and limitations" paragraph); also Dataset statistics Table 2 at line 353 ("dominated by two E3 ligases, reflecting the current state of PROTAC research").]

Minor 2: The lack of external validation is an important limitation.

Response: Now stated as Limitation item 6:

> *"No external validation: PROTAC-8K is the only publicly available benchmark with sufficient scale; generalization to data from other assay platforms or research groups is unvalidated."*

We also note in the introduction (line 133) that limitations are "prominently acknowledged," and in the conclusion (lines 749–750) that results are "not statistically superior to gradient boosting."

[Paper reference: Limitation item 6 at line 737 ("No external validation"); Introduction line 133; Conclusion lines 749–750.]

Summary of All Changes Relevant to Reviewer Y2rf

Concern	**Revision (lines in `paper.tex`)**
Best seed over-emphasized (M1)	Abstract L65; Introduction L132–133; Table 1 L421–438; §4.2 L440–442; Conclusion L749–751
Target-unseen unstable (M2)	§4.3 L590–599 (seed sensitivity + 6-seed); Table 1 L432; Figure 4 L605; p-values L441, L507, L595
Lysine claim unsupported (M3)	§5 L719–726; Figure 7 L729; Limitation 7 L738
E3 claims too broad (M4)	Abstract L65; §4.3 L608–620; Table 5 L625; Figure 5 L644; Conclusion L749
Validation misaligned (M5)	§3.5 L314–318; L310; Limitation 4 L735
Dataset limits buried (M6)	§4 L412–417; Table 2 L353
No external validation (M7)	Limitation 6 L737; Introduction L133; Conclusion L749–750
New analyses added in revision:	
6-seed extended validation	Table 1 L432; §4.3 L593–599; Figure 4 L605
Calibration analysis	§4.6 L674–700; Table 6 L683
Per-E3 breakdown	Table 5 L625; Figure 5 L644
VHL root cause analysis	§4.3 L612–620
BRD4 case study	§4.5 L655–672
Lysine paradox figure	Figure 7 L729
Statistical significance	L441, L507, L595
Honest performance assessment	§5 L713–717